

Question

Which document, adopted in 1947 in the aftermath of World War II, is considered foundational for modern medical ethics and, by extension, for the ethical treatment of human subjects in research that informs AI ethics today?

- A. The Belmont Report.
- B. The Nuremberg Code.
- C. The Declaration of Helsinki.
- D. The Declaration of Geneva.

Answer: B

Explanation

The Nuremberg Code (1947) established principles for ethical research involving human subjects, including voluntary consent. The Belmont Report came later (1979); the Declaration of Helsinki (1964) and Geneva (1948) built on the Nuremberg framework.

Question

Which of the following is NOT one of the five commonly cited AI ethics principles in the GARP Risk and AI curriculum?

- A. Nonmaleficence.
- B. Beneficence.
- C. Profit maximisation.
- D. Autonomy.

Answer: C

Explanation

The five principles are nonmaleficence, beneficence, justice, autonomy, and explainability. Profit maximisation is a business objective, not an ethical principle.

Question

Which of the following best limits the ethical duty of beneficence?

- A. There are no limits; beneficence is absolute.
- B. Scarce resources, competing obligations, and the constraint that one cannot alleviate all suffering in the world.
- C. Legal constraints only.
- D. Financial constraints only.

Answer: B**Explanation**

Beneficence is constrained by reasonableness: finite resources, competing moral demands, and the impossibility of universal assistance. Options A, C, and D are either too strong or partial.

Question

When a patient cannot make autonomous decisions (e.g., unconscious), who should make decisions on their behalf?

- A. The AI system.
- B. A surrogate decision-maker (family member, guardian) acting in the patient's best interests.
- C. The hospital's CEO.
- D. No one; decisions should be postponed indefinitely.

Answer: B

Explanation

When autonomy cannot be exercised directly, surrogates act in the patient's best interests. Options A, C, and D are not ethical or practical.

Question

A fintech company collects extensive personal data on its users because it might be useful someday. Which GDPR principle does this violate?

- A. Data minimisation — only data necessary for specified purposes should be collected.
- B. Right to access.
- C. Right to be forgotten.
- D. Data portability.

Answer: A

Explanation

GDPR's data minimisation principle requires that personal data be adequate, relevant, and limited to what is necessary. Options B, C, and D are other GDPR rights unrelated to the collection issue.

Question

Which of the following correctly distinguishes the CCPA from the GDPR?

- A. CCPA applies only to California residents; GDPR applies to all EU residents and has extraterritorial reach.
- B. CCPA covers all US residents.
- C. GDPR is voluntary.
- D. CCPA has no opt-out rights.

Answer: A

Explanation

CCPA is a California state law protecting California residents. GDPR applies across the EU (and to non-EU entities serving EU data subjects). Options B, C, and D are factually incorrect.

Question

A bank's board asks the CRO to design an ethics framework for the bank's AI use in lending. Which of the following is the most appropriate first step?

- A. Purchase an AI auditing tool.
- B. Define the bank's AI ethics principles (fairness, transparency, accountability) and how they align with existing risk appetite and regulatory expectations.
- C. Fire the data science team.
- D. Adopt all AI techniques without restriction.

Answer: B

Explanation

Ethics frameworks start with principles aligned to business and regulatory context; tools come later. Options A, C, and D are premature or incorrect.

Question

Which statement best captures the difference between transparency and interpretability?

- A. They are the same.
- B. Transparency is about disclosure and openness (what the model does, what data it uses); interpretability is about how understandable the model's internal mechanisms are.
- C. Transparency is only for regulators; interpretability is only for engineers.
- D. Interpretability is a legal requirement.

Answer: B**Explanation**

Transparency = openness about design, data, and decisions. Interpretability = degree to which internal logic is understandable. Options A, C, and D misstate the distinction.

Question

Which of the following is a recognised governance challenge for AI ethics?

- A. Power asymmetries between large tech firms and regulators.
- B. Lack of interpretability.
- C. Unpredictability of AI systems.
- D. All of the above.

Answer: D

Explanation

All three are recognised governance challenges. Power asymmetries, opacity, and unpredictability complicate regulation and oversight.

Question

Compared with medical ethics, which structural element is generally missing in AI ethics?

- A. Ethical codes.
- B. Ethics committees.
- C. Professional licensing and mandatory training.
- D. All of the above are more mature in medicine than in AI.

Answer: D

Explanation

Medical ethics has mature codes, ethics committees, licensing, and mandatory training. AI ethics lacks comparable institutional structures. Options A, B, and C are each missing or less developed.

Question

The EU AI Act came into force in 2024. Which statement about its scope is correct?

- A. It regulates only medical AI.
- B. It regulates AI systems based on a risk-tiered approach: unacceptable, high, limited, and minimal risk categories, with different obligations for each.
- C. It bans all AI.
- D. It applies only to EU-based entities.

Answer: B**Explanation**

The AI Act uses a risk-tiered approach. Prohibited practices are banned; high-risk systems face strict obligations; limited-risk systems have transparency requirements; minimal-risk systems are mostly unregulated. Options A, C, and D misstate the Act.

Question

Which of the following best describes the current US federal regulatory landscape for AI?

- A. A single comprehensive AI law.
- B. No federal AI law; a patchwork of state laws, executive orders, and agency guidance, plus frameworks like the NIST AI RMF.
- C. Only state-level AI regulation.
- D. Complete prohibition of AI use in finance.

Answer: B**Explanation**

As of the curriculum date, the US has no comprehensive federal AI law. Regulation is a patchwork of state laws (e.g., CCPA), executive orders, and agency frameworks (NIST AI RMF). Options A, C, and D are inaccurate.

Question

Which of the following best describes China's approach to AI regulation?

- A. China has no AI regulation.
- B. China has developed targeted regulations, including the Provisional Administrative Measures for Generative AI Services, the Personal Information Protection Law (PIPL), and algorithmic recommendation rules.
- C. China only regulates medical AI.
- D. China relies entirely on self-regulation by industry.

Answer: B

Explanation

China has adopted several targeted AI regulations, including the Generative AI Measures and PIPL. Options A, C, and D are factually incorrect.

Question

Which statement best captures the relationship between ethics and law in AI governance?

- A. Ethics and law are identical.
- B. Law establishes minimum standards of behaviour; ethics goes beyond law by guiding what is morally appropriate even when not legally required.
- C. Ethics is subordinate to law.
- D. Ethics replaces law.

Answer: B

Explanation

Ethics is broader than law: it reflects moral values and shapes laws but also guides behaviour where laws are silent. Options A, C, and D are incorrect.

Question

Which of the following best defines data governance in the AI/ML context?

- A. A system of decision rights and accountabilities for information-related processes.
- B. Data cleaning.
- C. Data storage.
- D. Data visualisation.

Answer: A

Explanation

Data governance is a system of decision rights and accountabilities, defining who can take what actions with what information, when, and how. Options B, C, and D are components, not the definition.

Question

A bank plans to use alternative data (e.g., social media, utility payments) to broaden credit access. Which of the following is the most important governance consideration before proceeding?

- A. Whether the data is cheap.
- B. Whether the bank has legal rights to use the data, has obtained appropriate consents, and has assessed privacy, fairness, and ethical implications.
- C. Whether the data is stored in the cloud.
- D. Whether the data analyst likes the data.

Answer: B**Explanation**

Alternative data requires heightened governance: legal rights (provenance), consent, privacy, fairness, and ethical review are essential. Options A, C, and D are secondary or irrelevant.

Question

Which of the following is a critical concern when using alternative data for AI/ML models?

- A. Alternative data is always more accurate than traditional data.
- B. Alternative data may be unverified, subject to sudden definition changes, or may disappear, requiring continuous monitoring.
- C. Alternative data never requires governance.
- D. Alternative data is always free of bias.

Answer: B

Explanation

Alternative data quality can shift rapidly: providers may change definitions, cease data feeds, or introduce biases. Continuous monitoring is essential. Options A, C, and D are false.

Question

Which of the following best describes data provenance in AI governance?

- A. The storage location of data.
- B. The origin and history of data, including its legal ownership and consent status.
- C. The processing speed of data.
- D. The file format of data.

Answer: B

Explanation

Provenance tracks the origin, custody, and legal right to use data. Options A, C, and D are unrelated.

Question

A firm classifies its data into Public, Internal, Confidential, and Restricted categories. What is the primary purpose of this classification?

- A. To increase storage costs.
- B. To determine handling, access controls, and protection levels appropriate to sensitivity and structure.
- C. To confuse employees.
- D. To comply with marketing requirements.

Answer: B

Explanation

Classification determines handling and protection based on sensitivity. Options A, C, and D are incorrect or nonsensical.

Question

Which of the following best describes the role of metadata in AI governance?

- A. Metadata is the raw data.
- B. Metadata is data about data (origin, structure, definitions, relationships), supporting discovery, lineage, and governance.
- C. Metadata is encrypted data.
- D. Metadata is synthetic data.

Answer: B

Explanation

Metadata provides context for data assets, enabling lineage tracking, discovery, and governance. Options A, C, and D misstate the concept.

Question

Which access control model assigns permissions based on the user's role within the organisation?

- A. Role-Based Access Control (RBAC).
- B. Attribute-Based Access Control (ABAC).
- C. Mandatory Access Control (MAC).
- D. Discretionary Access Control (DAC).

Answer: A

Explanation

RBAC assigns permissions based on roles. ABAC uses attributes, MAC uses security labels, DAC lets owners set permissions. Option A is the correct answer.

Question

Which of the following is the primary reason for encrypting sensitive data at rest?

- A. To increase storage capacity.
- B. To ensure confidentiality and integrity by preventing unauthorised access even if the physical media is compromised.
- C. To speed up queries.
- D. To comply with advertising standards.

Answer: B

Explanation

Encryption ensures confidentiality and integrity, protecting data even if physical media is compromised. Options A, C, and D are not the primary purpose.

Question

In a data governance framework, which role typically provides overall oversight of the data strategy, approves policies, and ensures alignment with business objectives?

- A. Data steward.
- B. Data custodian.
- C. Data governance committee or board-level function.
- D. Data entry clerk.

Answer: C

Explanation

The data governance committee or board-level function provides oversight and approves policies. Data stewards handle day-to-day data quality; custodians handle storage and technical controls. Options A, B, and D are operational roles.

Question

Which of the following is typically included in model governance policies?

- A. Definitions of a model and non-model.
- B. Risk tiering methodology.
- C. Documentation and validation requirements.
- D. All of the above.

Answer: D

Explanation

Model governance policies define what constitutes a model, how to risk-rank it, and what documentation, validation, and monitoring are required. All three components are essential.

Question

Which of the following is the primary purpose of model documentation in AI governance?

- A. To provide a record of the model's design, assumptions, limitations, and usage, supporting validation, audit, and ongoing governance.
- B. To increase model accuracy.
- C. To reduce training time.
- D. To eliminate the need for validation.

Answer: A

Explanation

Documentation supports validation, audit, reproducibility, and governance. It does not improve accuracy or eliminate validation. Options B, C, and D are incorrect.

Question

Which of the following best describes the purpose of a model inventory?

- A. To store the model's source code.
- B. To maintain a registry of all models, their owners, risk tiers, validation status, and interdependencies, enabling enterprise-level oversight.
- C. To train new models.
- D. To visualise model performance.

Answer: B

Explanation

The model inventory is a registry for governance, tracking models across the organisation. Options A, C, and D are unrelated or partial.

Question

What is a model landscape, and why is it important?

- A. A landscape is a list of models. It is not important.
- B. A landscape is a graphical or list-based representation of the interactions and interdependencies between models, revealing critical models and cascading risks.
- C. A landscape is a data pipeline.
- D. A landscape is a visualisation of model outputs.

Answer: B

Explanation

The model landscape maps interdependencies between models, revealing how failures can cascade. Options A, C, and D misstate the concept.

Question

Which of the following best describes the goal of model risk management (MRM)?

- A. To eliminate all model risk.
- B. To identify, assess, and mitigate model risks across the model lifecycle, balancing performance with governance and control.
- C. To maximise model complexity.
- D. To outsource model development.

Answer: B

Explanation

MRM aims to identify, assess, and mitigate model risk. It cannot eliminate risk entirely. Options A, C, and D are incorrect.

Question

In the Three Lines of Defence model applied to AI/ML models, what are the primary responsibilities of each line?

- A. First line: independent validation; second line: model development; third line: audit.
- B. First line: model owners and developers (build, operate); second line: independent risk management/validation (challenge, validate); third line: internal audit (assure).
- C. All three lines do the same thing.
- D. First line: audit; second line: governance; third line: development.

Answer: B

Explanation

First line owns models; second line independently validates and challenges; third line audits. Options A, C, and D misstate the roles.

Question

Who should typically own a model in an organisation's model governance framework?

- A. The vendor who built the model.
- B. A designated individual or committee in the business unit that uses the model and is accountable for its performance and use.
- C. The IT department only.
- D. The regulator.

Answer: B

Explanation

Model owners are typically in the business unit using the model, ensuring accountability. Options A, C, and D are not typical.

Question

When must model validation occur, and who should perform it?

- A. Only after a model has caused a loss; performed by the developers.
- B. Before initial deployment, periodically, and after material changes; performed by an independent validation function not involved in model development.
- C. Only annually; performed by the same team that developed the model.
- D. Never; validation is optional.

Answer: B

Explanation

Validation occurs pre-deployment, periodically, and after material changes, and must be independent. Options A, C, and D violate governance principles.

Question

What is the use test in model validation?

- A. A quantitative metric of model accuracy.
- B. A qualitative assessment of whether the model is being used appropriately for its intended purpose, considering human interaction, interpretation, and organisational context.
- C. A test of the model's execution speed.
- D. A test of the model's code.

Answer: B

Explanation

The use test evaluates whether the model is used correctly and appropriately. It is qualitative and focused on human and organisational factors. Options A, C, and D are distinct.

Question

What is the difference between white-box and black-box testing?

- A. White-box tests inputs and outputs only; black-box tests internal logic.
- B. White-box tests internal logic and code with full access; black-box tests only inputs and outputs without access to internals.
- C. They are identical.
- D. White-box is for AI; black-box is for statistics.

Answer: B

Explanation

White-box testing examines internal code and logic; black-box testing assesses behaviour based on inputs/outputs. Options A, C, and D are incorrect.

Question

Which of the following is the best practice for limiting model use?

- A. Allow any use case without restriction.
- B. Document approved uses and restrictions in the model inventory, and prevent use outside the validated domain (e.g., pricing one instrument with a model validated for another).
- C. Restrict all models to a single use case regardless of validation.
- D. Ignore model use restrictions.

Answer: B

Explanation

Model use limits should be documented and enforced to prevent misuse. Options A, C, and D violate governance best practice.

Question

When a model undergoes a material change (e.g., new data source, methodology update), what should happen in a well-governed framework?

- A. The change is implemented immediately without review.
- B. The change goes through a change-management process: impact assessment, documentation, validation, and approval.
- C. The model is retired.
- D. No action is needed.

Answer: B

Explanation

Material changes require impact assessment, documentation, validation, and approval. Options A, C, and D violate governance principles.

Question

How frequently should AI/ML models be reviewed?

- A. Never after initial deployment.
- B. Only when performance degrades.
- C. At least annually or more frequently if risk tier, usage, or environment warrants it; AI/ML models often require more frequent monitoring than traditional models.
- D. Only every five years.

Answer: C

Explanation

Periodic review is required, with frequency based on risk tier. AI/ML models often warrant more frequent review due to complexity and data drift. Options A, B, and D are incorrect.

Question

Which of the following best describes a key difference between reviewing AI/ML models and reviewing traditional statistical models?

- A. AI/ML models never require review.
- B. AI/ML models often need more frequent monitoring due to higher complexity, data drift, and evolving behaviour; validation focus shifts toward explainability and outcome testing rather than code inspection.
- C. Traditional models require more review.
- D. AI/ML models are simpler.

Answer: B**Explanation**

AI/ML models typically require more frequent monitoring and a shift in validation emphasis toward explainability, benchmarking, and outcome testing. Options A, C, and D are incorrect.

Question

Who sets the model risk appetite for a financial institution?

- A. The CRO alone.
- B. The board of directors.
- C. The model developers.
- D. External auditors.

Answer: B

Explanation

The board sets the model risk appetite; the model risk function maintains policies; developers build models; auditors independently check. Option B is correct.

Question

Which of the following is NOT a model implementation task?

- A. Model design.
- B. Core implementation.
- C. System implementation.
- D. Independent validation.

Answer: D

Explanation

Implementation tasks include model design, core implementation, and system implementation. Validation is separate. Options A, B, and C are implementation tasks.

Question

Core adaptation differs from system adaptation in that:

- A. Core adaptation modifies the model core (algorithm, parameters); system adaptation modifies the surrounding system (UI, data pipelines).
- B. They are the same.
- C. System adaptation is only for AI.
- D. Core adaptation is only for statistical models.

Answer: A

Explanation

Core adaptation changes the model itself; system adaptation changes the supporting environment. Options B, C, and D misstate the distinction.

Question

What should happen when a model is retired?

- A. It is silently deleted.
- B. It is formally decommissioned: reason for retirement documented, replacement model transitioned in, inventory updated, and documentation archived.
- C. It continues to be used without governance.
- D. Nothing; retirement is not a governance concept.

Answer: B

Explanation

Retirement is a formal process requiring documentation, transition, and inventory updates. Options A, C, and D violate governance.

Question

What should model risk reporting to senior management and the board typically include?

- A. Only the number of models in production.
- B. A summary of the model inventory, validation status, key risk issues, performance metrics, and remediation plans.
- C. The raw source code.
- D. Marketing material.

Answer: B

Explanation

Model risk reporting covers inventory, validation, risk issues, performance, and remediation. Options A, C, and D are incomplete or inappropriate.

Question

Which of the following is a key component of model change management?

- A. Informal verbal approval.
- B. Structured change requests, impact assessment, documentation, validation, and approval.
- C. Skipping validation for minor changes.
- D. Never changing models.

Answer: B

Explanation

Change management includes structured processes: request, impact assessment, documentation, validation, approval. Options A, C, and D violate governance.

Question

A complete data governance framework typically includes:

- A. Data strategy, quality, provenance, classification, metadata management, protection, access, compliance, and roles.
- B. Only data storage.
- C. Only data visualisation.
- D. Only data cleaning.

Answer: A

Explanation

Comprehensive data governance covers strategy, quality, provenance, classification, metadata, protection, access, compliance, and roles. Options B, C, and D are partial.

Question

Under GDPR, how quickly must an organisation notify the supervisory authority of a personal data breach?

- A. Immediately.
- B. Within 72 hours of becoming aware of the breach, unless the breach is unlikely to result in a risk to rights and freedoms.
- C. Within 30 days.
- D. Only if the breach is publicly reported.

Answer: B

Explanation

GDPR Art. 33 requires notification within 72 hours. Options A, C, and D are inaccurate.

Question

What is the board's role in data governance?

- A. Operational data entry.
- B. Approval of the data governance framework and oversight of compliance, risk management, and alignment with strategic objectives.
- C. Writing data-processing code.
- D. Selecting cloud vendors.

Answer: B

Explanation

The board approves the framework and provides oversight. Options A, C, and D are operational roles.

Question

Why is data provenance particularly important for AI training data?

- A. To speed up model training.
- B. To verify the legal right to use the data and to protect against legal claims (e.g., copyright, privacy violations) that could require the model to be destroyed.
- C. To increase storage requirements.
- D. To reduce training costs.

Answer: B

Explanation

Provenance verifies the legal right to use data, protecting against legal risk. Options A, C, and D are secondary or incorrect.

Question

Which of the following is a typical data classification level?

- A. Public, Internal, Confidential, Restricted.
- B. Fast, Slow, Medium.
- C. Red, Blue, Green only.
- D. Alpha, Beta, Gamma.

Answer: A

Explanation

Common classification levels are public, internal, confidential, and restricted. Options B, C, and D are not standard classification levels.

Question

Which of the following is a key advantage of adopting an AI ethics framework?

- A. It eliminates all AI risk.
- B. It provides structured guidance for ethical decisions, enhances trust, and reduces regulatory non-compliance risk.
- C. It maximises short-term profits.
- D. It replaces the need for model validation.

Answer: B

Explanation

Frameworks provide guidance, build trust, and reduce regulatory risk. Options A, C, and D are incorrect.

Question

Which of the following best captures the concept of Responsible AI?

- A. Maximising model accuracy regardless of consequences.
- B. Designing, developing, and deploying AI systems that are ethical, transparent, accountable, and aligned with societal values, while mitigating potential risks.
- C. Deploying AI as fast as possible.
- D. Ignoring regulatory requirements.

Answer: B

Explanation

Responsible AI encompasses ethics, transparency, accountability, and risk mitigation. Options A, C, and D are contrary to the concept.

Question

An insurance company uses a gradient-boosted model with hundreds of trees to price a complex commercial policy. The pricing actuary cannot fully explain how the model derives any particular premium. The risk function is concerned about “inscrutability”. Which mitigation strategy is most aligned with the GARP Risk and AI curriculum?

- A. Replace the model with a simple linear formula regardless of performance.
- B. Keep the model but implement post-hoc explainability tools (SHAP, LIME), benchmark against simpler interpretable challenger models, and establish ongoing monitoring and documentation.
- C. Deploy without any mitigation.
- D. Assume inscrutability is acceptable.

Answer: B

Explanation

Inscrutability is best mitigated by explainability tools, benchmarking, and governance, rather than replacing models. Options A, C, and D are impractical or unsafe.

Question

A team building a forecasting model uses standard random k-fold cross-validation on 10 years of monthly economic data. The cross-validation performance is excellent, but the deployed model performs poorly. What is the most likely reason?

- A. The model is underfit.
- B. Standard k-fold shuffles data, allowing future information to leak into training. Time-series-aware validation (rolling/expanding window) should be used instead.
- C. The data is corrupted.
- D. The model is too simple.

Answer: B

Explanation

Time-series requires temporal validation to avoid leakage. Options A, C, and D are unlikely explanations.

Question

Which of the following is a qualified analytical tool (non-model) rather than a model under SR 11-7?

- A. A Monte Carlo simulation for portfolio VaR.
- B. A logistic regression for credit scoring.
- C. A simple deterministic spreadsheet performing a fixed calculation.
- D. A neural network.

Answer: C

Explanation

Non-models are tools with no material uncertainty or judgment (fixed calculations). Options A, B, and D are models with statistical uncertainty.

Question

Which of the following is a standard factor in model risk ranking?

- A. Materiality of model use.
- B. Complexity of methodology.
- C. Potential exposure (financial, regulatory, reputational).
- D. All of the above.

Answer: D

Explanation

Model risk ranking considers materiality, complexity, and exposure. All three are standard factors.

Question

Why must the model risk function be independent from model developers?

- A. To ensure objective challenge of model assumptions and results, preventing conflicts of interest.
- B. Because regulators require it for tax purposes.
- C. To reduce costs.
- D. To speed up validation.

Answer: A

Explanation

Independence ensures objective validation and challenge. Options B, C, and D are not the primary reason.

Question

A data scientist uploads proprietary customer data to a third-party LLM platform to help debug code. Which of the following is the most serious governance concern?

- A. The LLM might return incorrect code.
- B. Uploading proprietary or personally identifiable data to a third-party platform without governance approval violates data protection and confidentiality policies, potentially leading to regulatory and reputational harm.
- C. The tool might be slow.
- D. The tool might charge fees.

Answer: B

Explanation

Proprietary or personal data must not be shared with third parties without governance approval and legal basis. Options A, C, and D are secondary.

Question

What is the purpose of outcomes analysis in model validation?

- A. To analyse model outputs against actual outcomes, checking that the model's predictions correspond to realised business or risk outcomes.
- B. To measure training time.
- C. To test the model's code line by line.
- D. To compare model speed across versions.

Answer: A

Explanation

Outcomes analysis checks that model predictions align with reality. Options B, C, and D are unrelated.

Question

Which of the following best describes a model risk appetite statement?

- A. A marketing statement.
- B. A statement defining the level of model risk the organisation is willing to accept, guiding model governance and validation activities.
- C. A statement about IT infrastructure.
- D. A statement about employee salaries.

Answer: B

Explanation

The model risk appetite defines acceptable risk levels and guides governance. Options A, C, and D are unrelated.

Question

Which of the following is a GenAI-specific governance challenge beyond traditional ML?

- A. Open-ended, unstructured output.
- B. Non-determinism of model outputs.
- C. Vendor dependence and rapid model updates.
- D. All of the above.

Answer: D

Explanation

GenAI adds open-ended output, non-determinism, and vendor dependence as governance challenges. All three are material.

Question

Why is continuous monitoring particularly important for GenAI systems?

- A. Because GenAI models do not require monitoring.
- B. Because outputs are non-deterministic, providers may update models, and drift can be subtle, requiring ongoing oversight.
- C. Because GenAI systems are cheap to train.
- D. Because GenAI systems require no validation.

Answer: B

Explanation

GenAI's non-determinism, vendor updates, and drift require continuous monitoring. Options A, C, and D are incorrect.

Question

What is the purpose of human-in-the-loop (HITL) design in AI systems?

- A. To speed up AI processing.
- B. To include human review and verification at key decision points, mitigating risks of AI errors and ensuring appropriate oversight.
- C. To eliminate human roles.
- D. To replace model validation.

Answer: B

Explanation

HITL inserts human oversight at critical points, mitigating AI risk. Options A, C, and D are contrary to the concept.

Question

Which of the following should be documented in model documentation?

- A. Purpose, methodology, data, assumptions, limitations, and validation results.
- B. Only the model's source code.
- C. Only marketing materials.
- D. Only the model's accuracy.

Answer: A

Explanation

Model documentation is comprehensive: purpose, methodology, data, assumptions, limitations, validation. Options B, C, and D are incomplete.

Question

Which of the following best describes the purpose of model monitoring?

- A. To detect performance degradation, data drift, and concept drift, enabling timely intervention (retraining, recalibration, retirement).
- B. To increase model complexity.
- C. To eliminate the need for validation.
- D. To market the model.

Answer: A

Explanation

Monitoring detects degradation and drift. Options B, C, and D are unrelated.

Question

What is the difference between data drift and concept drift?

- A. They are identical.
- B. Data drift is a change in the distribution of input features; concept drift is a change in the relationship between features and the target variable.
- C. Data drift only affects regression models.
- D. Concept drift only affects classification models.

Answer: B

Explanation

Data drift = input distribution shift; concept drift = feature-target relationship shift. Options A, C, and D misstate the concepts.

Question

Which of the following is a valid reason for retiring a model?

- A. The model no longer serves a business need.
- B. A better model is available.
- C. Regulatory or business changes render the model obsolete.
- D. All of the above.

Answer: D

Explanation

Models are retired when needs change, better models exist, or regulation/business makes them obsolete. All three are valid reasons.

Question

What is adversarial testing in the context of AI/ML models?

- A. Normal unit testing.
- B. Testing the model's response to deliberately crafted inputs (adversarial examples) designed to cause incorrect or unsafe behaviour.
- C. Marketing the model.
- D. Documenting the model.

Answer: B

Explanation

Adversarial testing probes for robustness by using crafted inputs. Options A, C, and D are unrelated.

Question

Why is identifying model interdependencies important in a model landscape?

- A. To reduce model count.
- B. Because one model's outputs may be another's inputs; a failure in one can cascade through others, creating systemic risk.
- C. To speed up training.
- D. To reduce costs.

Answer: B

Explanation

Interdependencies create cascading risks, which must be governed. Options A, C, and D are not the primary concern.

Question

What is the typical role of a data governance committee?

- A. Operational data entry.
- B. Oversight of data strategy, policies, and practices, without direct operational responsibility.
- C. Building models.
- D. Marketing data products.

Answer: B

Explanation

The data governance committee provides oversight, not operational work. Options A, C, and D are operational roles.

Question

Which of the following best distinguishes global from local explainability?

- A. Global = overall model behaviour; local = specific prediction.
- B. Global = specific prediction; local = overall behaviour.
- C. They are identical.
- D. Local is for tree models only.

Answer: A

Explanation

Global = overall; local = per-prediction. Options B, C, and D misstate the distinction.

Question

Which of the following is NOT typically a source of model risk?

- A. Poor data quality.
- B. Inappropriate model assumptions.
- C. Implementation errors.
- D. Increasing model training time.

Answer: D

Explanation

Training time is a performance concern, not a source of model risk. Poor data, wrong assumptions, and implementation errors are all recognised sources.

Question

Which of the following is a governance concern when using a third-party GenAI model?

- A. The vendor may update the model without notice, changing behaviour unpredictably.
- B. The vendor may use inputs for training.
- C. The vendor's terms of service may limit liability.
- D. All of the above.

Answer: D

Explanation

Vendor model updates, input handling, and contractual terms are all governance concerns. All three are material.

Question

Which of the following best describes the purpose of data classification in a governance framework?

- A. To categorise data by sensitivity and structure so that appropriate handling, access, and protection controls can be applied.
- B. To increase data storage costs.
- C. To reduce data volume.
- D. To remove data.

Answer: A

Explanation

Classification drives controls. Options B, C, and D are incorrect.

Question

Which of the following is NOT a recognised source of algorithmic bias?

- A. Problem specification and feature selection.
- B. Data collection and composition.
- C. Model development choices.
- D. Increasing the number of training epochs.

Answer: D

Explanation

Training epochs affect convergence, not fundamental bias sources. Sources of bias are problem specification, data, model development, and deployment.

Question

When base rates differ between groups, can all fairness metrics (demographic parity, predictive rate parity, equal opportunity) be satisfied simultaneously?

- A. Yes, always.
- B. No, they are mathematically incompatible under unequal base rates.
- C. Only for logistic regression.
- D. Only for tree models.

Answer: B

Explanation

The impossibility theorem (Kleinberg, Chouldechova) shows these cannot all hold under unequal base rates. Options A, C, and D are incorrect.

Question

Which of the following is a symptom of overfitting?

- A. Low training error, high test error.
- B. High training error, low test error.
- C. Both training and test error high.
- D. Both training and test error low.

Answer: A

Explanation

Overfitting: fits noise on training, fails on test. Options B, C, and D do not describe overfitting.

Question

Which of the following is a symptom of underfitting?

- A. Low training error, high test error.
- B. High training error, high test error.
- C. Only training error high.
- D. Only test error high.

Answer: B

Explanation

Underfitting: model cannot capture underlying pattern; both errors high. Options A, C, and D are incorrect.

Question

Which of the following is the most effective mitigation for LLM hallucinations in a high-stakes risk-management application?

- A. Increase temperature.
- B. Use retrieval-augmented generation (RAG) to ground outputs in authoritative sources, and add human verification for critical decisions.
- C. Use a smaller model.
- D. Ignore hallucinations.

Answer: B

Explanation

RAG grounds model responses in retrieved content; human verification catches remaining errors. Options A, C, and D are inadequate or incorrect.

Question

Which of the following best describes the relationship between data quality and model risk?

- A. Data quality is independent of model risk.
- B. Poor data quality (inaccuracy, inconsistency, missingness) is a major source of model risk; garbage in, garbage out.
- C. Data quality only matters for regression models.
- D. Data quality only matters for neural networks.

Answer: B

Explanation

Data quality directly affects model reliability; poor data is a primary source of model risk. Options A, C, and D are incorrect.

Question

Which of the following best describes the risk of AI overreliance?

- A. A minor issue.
- B. A substantial risk where users defer excessively to AI recommendations, reducing critical judgment and potentially amplifying errors (automation bias).
- C. A benefit of AI.
- D. Only relevant to unsupervised models.

Answer: B

Explanation

Overreliance (automation bias) is a recognised AI risk. Options A, C, and D misstate the concept.

Question

Which of the following best describes why AI/ML models should be included in a model inventory?

- A. To ensure they are governed under the same framework as traditional models, with appropriate risk ranking, validation, and monitoring.
- B. Because regulators require all IT systems to be inventoried.
- C. Because they are cheaper to build.
- D. Because they are always high-risk.

Answer: A

Explanation

AI/ML models require governance just like traditional models. Options B, C, and D are incorrect.

Question

An organisation has 50 models; outputs of 10 feed into inputs of 20 others. Which governance artefact is designed to capture these interactions?

- A. Model inventory.
- B. Model landscape.
- C. Data dictionary.
- D. Validation report.

Answer: B

Explanation

The model landscape captures interdependencies. The inventory lists models; the data dictionary describes data; the validation report documents validation.

Question

A vendor provides a model trained on data of unclear provenance. Which governance issue is most relevant?

- A. Model complexity.
- B. Legal rights and privacy violations, which could require the model to be withdrawn or destroyed.
- C. Model accuracy.
- D. Training time.

Answer: B

Explanation

Unclear provenance is a legal/governance risk; models built on unlawfully acquired data may need to be destroyed. Options A, C, and D are secondary.

Question

Which of the following factors most directly influences the required frequency of model validation?

- A. The vendor's preference.
- B. The risk tier of the model (higher risk = more frequent validation), usage volume, and rate of environmental change.
- C. The CEO's schedule.
- D. Marketing department requirements.

Answer: B

Explanation

Risk tier, usage, and environmental change drive frequency. Options A, C, and D are not relevant.

Question

Why is it important to enforce limits on model use?

- A. Because all uses are equivalent.
- B. Because using a model outside its validated scope (e.g., different products, geographies, or time horizons) can produce materially incorrect results and unexpected risks.
- C. Because regulators forbid all models.
- D. Because model use has no implications.

Answer: B

Explanation

Use limits prevent models from being applied outside their validated domain. Options A, C, and D are incorrect.

Question

Which of the following is a GenAI-specific risk not typically present in traditional statistical models?

- A. Overfitting.
- B. Hallucination — generation of plausible but factually incorrect output.
- C. Multicollinearity.
- D. Heteroskedasticity.

Answer: B

Explanation

Hallucination is unique to generative models. Options A, C, and D are traditional statistical risks.

Question

Which of the following is an internal governance lever for GenAI?

- A. Assigning clear ownership and accountability across the GenAI lifecycle.
- B. Tracking changes in the external regulatory environment.
- C. Relying on external standards only.
- D. Outsourcing all risk management.

Answer: A

Explanation

Internal levers include ownership, review processes, and escalation paths. Option B is external. Options C and D are not governance levers.

Question

Which of the following is an external governance constraint for GenAI?

- A. Evolving regulatory requirements (e.g., EU AI Act).
- B. Contractual terms with model providers.
- C. Emerging standards (e.g., NIST AI RMF, ISO/IEC 42001).
- D. All of the above.

Answer: D

Explanation

All three are external governance constraints. Regulatory, contractual, and standards-based constraints shape GenAI governance.

Question

Which of the following best captures why GenAI governance is harder than traditional ML governance?

- A. GenAI systems are opaque, non-deterministic, and often depend on external providers who may update models without notice.
- B. GenAI systems are simpler.
- C. GenAI systems have no regulatory exposure.
- D. GenAI systems are always open-source.

Answer: A

Explanation

Opacity, non-determinism, and vendor dependence are the key governance challenges. Options B, C, and D are incorrect.

Question

Which of the following is a practical benefit of an ethical AI framework in a financial institution?

- A. Improved ability to evaluate ethical dilemmas, increased trust, and reduced regulatory risk.
- B. Increased model accuracy.
- C. Reduced training time.
- D. Elimination of all AI risks.

Answer: A

Explanation

Ethics frameworks help evaluate dilemmas, build trust, and reduce regulatory risk. Options B, C, and D are not the primary benefits.

Question

Which of the following is a commonly used post-hoc explainability technique for complex models?

- A. Linear regression.
- B. SHAP values.
- C. Logistic regression.
- D. Ridge regression.

Answer: B

Explanation

SHAP is a post-hoc technique based on Shapley values. Options A, C, and D are modelling techniques, not explainability methods.

Question

A firm stores customer Social Security Numbers alongside publicly available firm financials. How should these be classified?

- A. Both public.
- B. SSNs: restricted/confidential; financials: public.
- C. Both restricted.
- D. Both internal.

Answer: B

Explanation

SSNs are highly sensitive (restricted/confidential). Public financials are public. Options A, C, and D misclassify.

Question

A data scientist needs to know the exact definition of a column "transactionamt" in a large database. Where is this information most likely stored?

- A. In the raw data.
- B. In metadata (data dictionary).
- C. In the model.
- D. In the audit log.

Answer: B

Explanation

Metadata (data dictionary) describes column definitions, types, and relationships. Options A, C, and D are unrelated.

Question

Which of the following is an example of a role-based access control decision?

- A. All employees can access all data.
- B. Only members of the credit risk team can access the credit risk database, based on their role.
- C. All employees can access personal data if they sign a form.
- D. Access is decided by seniority only.

Answer: B

Explanation

RBAC grants access based on roles. Options A, C, and D do not reflect RBAC principles.

Question

Which of the following is a typical model validation activity?

- A. Assessment of conceptual soundness.
- B. Review of data appropriateness.
- C. Review of implementation and outcomes analysis.
- D. All of the above.

Answer: D

Explanation

Validation covers conceptual soundness, data, implementation, and outcomes. All three are standard activities.

Question

Which of the following would most likely elevate an AI/ML model to a higher risk tier?

- A. High complexity and use in a regulatory capital calculation.
- B. Frequent use across business units with material financial impact.
- C. Use with alternative data sources introducing fairness and privacy concerns.
- D. All of the above.

Answer: D

Explanation

Complexity, materiality, and novel data sources all increase risk tier. All three elevate the ranking.

Question

Which of the following should a model risk report to the board typically include?

- A. The full model inventory, validation status, key risk issues, and remediation plans.
- B. The full source code of all models.
- C. Detailed marketing plans.
- D. Employee salaries.

Answer: A

Explanation

Reports include inventory, validation status, risk issues, and remediation. Options B, C, and D are inappropriate.

Question

A marketing team wants to collect more customer data than is strictly necessary for a new feature. What does the data minimisation principle advise?

- A. Collect as much data as possible to maximise future options.
- B. Collect only what is necessary for the specified purpose; unnecessary data should not be collected.
- C. Collect data without any governance.
- D. Ignore the principle.

Answer: B

Explanation

Data minimisation limits collection to what is necessary. Options A, C, and D violate the principle.

Question

Which of the following is NOT one of the widely cited Responsible AI principles?

- A. Fairness.
- B. Transparency.
- C. Accountability.
- D. Maximising short-term shareholder value regardless of harm.

Answer: D

Explanation

Responsible AI principles include fairness, transparency, accountability, privacy, and safety. Maximising short-term shareholder value at the expense of harm is contrary to Responsible AI.

Question

A financial institution is establishing an AI/ML governance framework. Which of the following is a key recommendation from the curriculum?

- A. Build a new framework entirely from scratch.
- B. Start from existing model governance frameworks and extend them with AI-specific considerations (data, explainability, ethics, fairness, and GenAI-specific governance).
- C. Focus only on technical performance.
- D. Ignore regulatory expectations.

Answer: B

Explanation

Extending existing governance frameworks with AI-specific elements is recommended. Options A, C, and D are not aligned with best practice.

Question

Which of the following best summarises the guiding principle of AI/ML governance in financial services?

- A. Maximise model complexity at all costs.
- B. Balance model performance with appropriate governance, validation, monitoring, fairness, explainability, and ethical considerations, ensuring alignment with regulatory requirements and business objectives.
- C. Deploy all models without validation.
- D. Ignore risks.

Answer: B

Explanation

Governance balances performance with validation, monitoring, fairness, ethics, and compliance. Options A, C, and D are contrary to the principle.